

Prompt: AI Strategy Overhaul (State & Decision Tree)

We are changing the structure of the AI strategy. Please remove all phase-based logic and replace it with a State-dependent decision tree. The AI must loop through this decision tree for each territory repeatedly until no more valid moves exist, then end its turn for that territory.

YOUR TASK: Update the AI strategy to match the following logic.

General Rules (Always Active):

- * A territory must never drop into a negative credit balance for the upcoming round after buying or upgrading units/buildings.
- * Border tiles facing enemy territories must not be left without at least a Strength 1 defense (unit or building), unless the AI is executing Priority 1 or Priority A.
- * Defensive buildings should be placed 1 hex from the border, if possible. If not, place it on a border tile.
- * Units should only move onto water/lake tiles if there is a valid land tile to conquer from there (use a BFS check).
- * Units moving onto water/lake must always bring enough credits to cover 2x their upkeep.

Round 1 - Free Tower Only:

- * Place the tower on the tile that covers the most AI-owned tiles with its Zone of Control (ZoC).
- * If there's a tie, pick the tile closest to the center of the board.

STATE Evaluation (Round 2 and onwards):

- * At the start of each territory's turn, the AI evaluates if its State is DEFENDING or ATTACKING.
- * **DEFENDING:** Only active if there is an enemy unit within 3 tiles of the border whose strength is strictly greater than the AI's strongest unit or building in that territory.
- * **ATTACKING:** The default state if not Defending.
- * If STATE = DEFENDING, evaluate the *entire* decision tree starting from Priority 1.
- * If STATE = ATTACKING, skip Priority 1 and 2, and evaluate directly from Priority A.

THE DECISION TREE: *(Evaluate for each valid action: moving an existing unit, merging 2 units and moving, or buying a new unit directly into an attack)*

Priority 1 - DEFENDING ONLY - Attack stronger unit's area:

- * **Condition:** Can the AI split the stronger enemy's territory into 2 parts?

- * **Step 1:** If yes, check if this split causes the stronger enemy unit to die at the start of its turn due to negative income. If yes, execute the move. If no, attack the tile that shrinks the stronger unit's resulting area as much as possible.
- * **Step 2:** If the initial condition is no, move to the next priority.

Priority 2 - DEFENDING ONLY - Defend against stronger unit:

- * **Condition:** Can the AI upgrade an existing building to match the stronger enemy's strength?
- * **Step 1:** If yes, execute the upgrade.
- * **Step 2:** If no, can the AI upgrade an existing unit within 5 tiles of the stronger enemy? If yes, do it and move the unit towards the stronger enemy.
- * **Step 3:** If no to Step 2, can the AI build a new building matching the stronger enemy's strength within 5 tiles of it? If yes, build it as close to the enemy as possible.
- * **Step 4:** If no to Step 3, can the AI build a building with 1 less strength than the stronger unit on a tile that is 1 step away from the enemy's border? If yes, build it.
- * **Step 5:** If no to all, move to the next priority.

Priority A - Attack area and split in 2:

- * **Condition:** Can the AI split an enemy territory into 2 parts?
- * **Step 1:** If yes, check if this split instantly kills a unit/building (1-hex rule) or kills them next turn via negative income. If yes, execute it. If no, attack the tile that minimizes the size of the resulting area containing the most enemy units and buildings.
- * **Step 2:** If the initial condition is no, move to the next priority.

Priority B - Connect own areas:

- * **Condition:** Can the AI connect two of its own separated territories via 1 or 2 middle tiles?
- * **Step 1:** If yes, execute it, prioritizing the connection of larger territories over smaller ones.
- * **Step 2:** If no, move to the next priority.

Priority C - Build defense:

- * **Condition:** Can the AI build a defensive building to protect currently undefended border tiles (no units/buildings) that border an enemy territory of 2+ tiles?
- * **Step 1:** If yes, build it.
- * **Step 2:** If no, move to the next priority.

Priority D - Build city:

- * **Condition:** Can the AI build a city?
- * **Step 1:** If yes, execute it, prioritizing placing it as far away from the largest enemy territories as possible while still being protected by a building's ZoC.
- * **Step 2:** If no, move to the next priority.

Priority E - Border expansion:

- * **Condition:** Can the AI conquer a neighboring tile occupied by an enemy unit or building?
- * **Step 1:** If yes, execute it, prioritizing the strongest units/buildings it can defeat.
- * **Step 2:** If no, can the AI conquer an empty enemy neighbor tile? If yes, execute it, prioritizing tiles from the largest neighboring territory.
- * **Step 3:** If no to Step 2, can the AI conquer an empty neutral neighbor tile? If yes, execute it, prioritizing City first, then Grass/Forest, then Desert.
- * **Step 4:** If no to all, move to the next priority.

Priority F - Clear rebels:

- * **Condition:** Can the AI move an existing unit onto a tile with a rebel?
- * **Step 1:** If yes, do it using the unit with the lowest strength.
- * **Step 2:** If no, can the AI build a Strength 1 unit directly on the rebel? If yes, build it.
- * **Step 3:** If no to all, move to the next priority.

Priority G - Move unit towards enemy:

- * **Condition:** Can the AI move an existing unit closer to an enemy territory where it's possible to conquer a tile?
- * **Step 1:** If yes, move it, but try to place it 1 tile away from the border if possible.
- * **Step 2:** If no, move to the next priority.

Priority H - Remove unnecessary defenses:

- * **Condition:** Can the AI remove one of its own defensive buildings located more than 5 tiles away from the nearest border to a 2+ tile enemy territory?
- * **Step 1:** If yes, remove it.
- * **Step 2:** If no, End turn.